On the Multiplicatively Closed Set Generated by 2 and 3

Partial results, an exact census up to 10^8, and a rejected mean-field heuristic

Research dossier for Ben Green's open problem GREEN-036 (100 Open Problems, Number Theory) - August 25, 2026.
Source: https://www.unsolvedmath.com/problems/GREEN-036 - status: **OPEN**

Abstract. Let A be the smallest set of positive integers containing 2 and 3 and closed under the operation (a1,a2) |-> a1*a2 - 1. Green's problem GREEN-036 asks whether A has positive asymptotic density. **The problem remains open.** This dossier contributes: (i) complete proofs of a well-foundedness structure theorem, of the modular obstruction (Lemma A: A avoids 1 mod 3, whence upper density d*(A) <= 2/3), and of an explicit floor((floor(log2 N)-1)^2/4) ~ 0.5203 (ln N)^2 lower bound (Lemma B); (ii) an exact, independently cross-checked census F(10^8) = |A ^ [1,10^8]| = 55,939,931 with per-decade statistics showing monotonically rising density 0.500 -> 0.559 and local power-law exponents descending to 1.0079; (iii) a mean-field self-consistency heuristic sum_{b in A} b^-(1+alpha) = 1 predicting alpha* = 0.4734, which is quantitatively rejected by the census (factor 9124), indicating that membership correlations, not pseudorandomness, govern A; and (iv) a residue census showing that among all moduli m <= 500 only m = 3 yields a proper closed residue set, so no improvement on 2/3 is known. Statements are labeled [PROVEN], [COMPUTED], or [HEURISTIC] throughout.

1. Problem statement and status

Let A be minimal with 2, 3 in A and a1, a2 in A implying a1*a2 - 1 in A. Question: does the asymptotic density d(A) = lim F(N)/N exist and satisfy d(A) > 0, where F(N) = |A ^ [1,N]|? The question is **open**; nothing here settles it. Sections 2-5 establish partial results with proofs, Section 6 reports an exact computational census, and Section 7 derives a heuristic together with its own empirical rejection.

2. Well-foundedness, exact characterization, and generation algorithm

Structural fact [PROVEN]. 1 is not in A (seeds are 2, 3; ab - 1 >= 2*2 - 1 = 3 for a,b >= 2), and whenever n = a1*a2 - 1 with a1,a2 in A one has a1,a2 >= 2 and hence a1, a2 <= (n+1)/2 < n. Values strictly decrease along any generation tree toward its leaves.

Theorem 1 (exact characterization) [PROVEN]. n is in A if and only if n is 2 or 3, or n = ab - 1 for some a,b in A. *Proof.* Minimality gives one direction directly: if n > max(a,b) equals ab - 1 with a,b in A then n cannot be excluded by any closed set containing the seeds, induction on max(a,b) being well-founded by the structural fact. Conversely the set {2,3} u {ab-1 : a,b in A} is contained in A by closure. ■

This makes A the least fixed point of S |-> S u {ab-1} and justifies the increasing-order dynamic program used by the verifier: membership bits in a bytearray; a min-heap of discovered-but-unexpanded values; a sorted partner list holding A ^ [1, sqrt(X)]. Each popped minimum m is expanded against partners a <= min(m, X//m) <= sqrt(X).

Theorem 2 (exactness) [PROVEN]. At termination the marked bytes are exactly A ^ [1,X]. *Proof.* Soundness is clear: inserted values ma - 1 use factors already certified to be in A. For completeness note pushes always exceed the popped value (ma - 1 >= 2m - 1 > m), so pops occur in increasing order; by strong induction on n, before any pop of value > n all elements of A ^ [1,n] are discovered (a non-seed n' <= n has witness pair with larger factor M <= (n'+1)/2 < n already discovered and hence expanded while the heap minimum was still below n, inserting n'). Every pair (a,b) with product <= X is consumed at the expansion of its larger factor M <= (X+1)/2, whose partner min(a,b) <= min(M, X/M) <= sqrt(X) is then present in the small list; so expansion against A ^ [1, sqrt(X)] misses nothing. ■

Cross-check [COMPUTED]. Independent naive fixpoint iteration reproduces the DP census exactly on [1,3000] (|A| = 1358 on both sides). Production run at X = 10^8: 6 min 31 s wall time, pure Python standard library.

3. Lemma A: the modular obstruction and the 2/3 ceiling

Lemma A [PROVEN]. No element of A is congruent to 1 modulo 3. *Proof.* Work in Z/3. The seed residues 2 and 0 both lie in S = {0,2}. For r,s in S: if r = 0 or s = 0 then rs - 1 = -1 = 2 (mod 3); if r = s = 2 then rs - 1 = 3 = 0 (mod 3). Hence S is closed under the operation and contains the seeds, so by minimality the residues of A lie in S: never 1 modulo 3. ■

Corollary [PROVEN]. Upper density d*(A) <= 2/3 = 0.666667: each admissible class carries at most N/3 + O(1) elements below N.

Census check [COMPUTED]. Zero of the 55,939,931 elements below 10^8 are 1 mod 3. Note A is a proper subset of the allowed classes: e.g. 6, 11, 62 are not in A although none is 1 mod 3 (their n+1 admit no divisor pair from A x A).

4. Residue sets modulo arbitrary moduli

Proposition 3 [PROVEN]. For every modulus m the residue set {a mod m : a in A} equals exactly the minimal closed set S_m generated from {2 mod m, 3 mod m}. *Proof.* Containment: reduction commutes with the operation, so generation trees reduce into any closed set containing the seed residues. Reverse: every element of S_m arises after finitely many closure steps from the seeds, and each step lifts back through actual elements of A by induction on the step count. ■ Consequently d*(A) <= |S_m|/m for every m, with CRT combination across coprime moduli.

modulus m	|S_m|	|S_m|/m	remark
3	2	0.666667	S_3 = {0,2}: Lemma A
9	6	0.666667	{0,2,3,5,6,8}: repackages mod 3
all other primes p <= 500 (94)	p	1	S_p = Z/p, full set
composite 4..500, m not in {9}	-	> 2/3	none beats 2/3

Table 1. Closed residue sets [COMPUTED]: 95 primes and all composite moduli up to 500 scanned. Only m = 3 is proper; the best proven ceiling remains 2/3 and the residue method is saturated at small heights.

5. Lemma B: an explicit logarithmic lower bound

Two families live in A [PROVEN]: P_k = 2^k + 1 (k >= 1), the orbit of 3 under x |-> 2x - 1 (since t - 1 doubles along the orbit), and V_j = (3^(j+1)+1)/2 (j >= 1), the orbit of 2 under x |-> 3x - 1 (fixed point 1/2: t_k - 1/2 = 3^k (2 - 1/2)). Closure produces all pairwise images.

Lemma B [PROVEN]. For N >= 8, F(N) >= floor((floor(log2 N) - 1)^2 / 4) >= (log2 N - 2)^2/4 = (ln N)^2/(4 ln^2 2) + O(ln N), with explicit constant c = 1/(4 ln^2 2) = 0.5203.

Proof. Consider z(i,j) = (2^(i+1)+1)(2^(j+1)+1) - 1 for integers 0 <= i <= j; all lie in A by two applications of closure, and z = 2^(i+j+2) + 2^(i+1) + 2^(j+1). **Distinctness.** If i < j then z = 2^(i+1) (1 + 2^(j-i) + 2^(j+1)) with odd bracket, so v2(z) = i+1 and odd part q = 1 + 2^(j-i) + 2^(j+1). If i = j >= 1 then z = 2^(i+2)(2^i+1): v2(z) = i+2, odd part 2^i + 1. The degenerate pair (0,0) gives z = 8 = 2^3, the only pure power of two in the family (every off-diagonal entry has odd part q >= 11 and value >= 14; every diagonal entry with i >= 1 has odd part 2^i + 1 > 1 and value >= 24). Suppose z(i,j) = z(i',j'). Off-diagonal vs off-diagonal: equal v2 forces i = i', and equality of odd parts reads 2^(j-i) (1 + 2^(i+1)) = 2^(j'-i) (1 + 2^(i+1)); cancelling the odd integer forces j = j'. Diagonal vs diagonal: equal v2 forces i = i'. Off-diagonal vs diagonal would force simultaneously i+1 = i'+2 and 1 + 2^d + 2^(d+i+1) = 2^i' + 1, i.e. 2^d (1 + 2^(i+1)) = 2^i' with d >= 1: impossible since a pure power of two cannot be divisible by the odd integer 1 + 2^(i+1) >= 3. Hence the map is injective. **Counting.** Let lambda = floor(log2 N); whenever s := i+j+3 <= lambda, z <= 2^(s-1) + 2^(s-2) + 2^(s-2) = 2^s with strict inequality (equality would force s = 2(s-2)+3), so z < 2^s <= N. The pairs with i <= j, i+j <= lambda-3 number sum_{s=0}^{lambda-3}(floor(s/2)+1) = floor((lambda-1)^2/4). ■

Mixed family [COMPUTED; injectivity unproven]. With U_i = 2^(i+1)+1 and V_j = (3^(j+1)+1)/2, the values w(i,j) = U_i V_j - 1 all lie in A; a full injectivity proof would raise the constant to 1/(2 ln 2 ln 3) = 0.6566 (ln N)^2.

The 2-adic cascade recovers min(i+1, v2(V_j - 1)) but ties block complete recovery. Computed up to 10^8: 216 values, zero collisions, all in A; the proven family contributes 168 values against the proven floor 156. Both bounds are exponentially far from the true F(10^8) = 55,939,931: their value is unconditional methodology, not approximation.

6. Exact census up to 10^8

k	F(10^k)	F(10^k)/10^k	local exp	harm incr	delta_hat
1	5	0.500000	-	1.2694	0.5513
2	39	0.390000	0.8921	0.8178	0.3552
3	422	0.422000	1.0342	0.9470	0.4113
4	4,805	0.480500	1.0564	1.1029	0.4790
5	51,508	0.515080	1.0302	1.1838	0.5141
6	535,585	0.535585	1.0170	1.2318	0.5350
7	5,493,428	0.549343	1.0110	1.2637	0.5488
8	55,939,931	0.559399	1.0079	1.2871	0.5590

Table 2. Exact per-decade census [COMPUTED]. local exp = log(F(10^k)/F(10^(k-1)))/log 10; harm incr = sum of 1/b over A ^ (10^(k-1), 10^k]; delta_hat = incr / ln 10 estimates density.

Reading of the data [COMPUTED trend; OPEN conclusion]: raw density rises in every decade; local exponents descend toward 1 from above (top value 1.0079); per-decade harmonic mass approaches delta_hat ln 10 with delta_hat climbing through 0.56. A log-log least-squares fit over decades 4-8 gives alpha_OLS = 1.01600 +- 0.00254 with R^2 = 0.99998, consistent with near-linear growth F(N) ~ c_N N and drifting density. All trends are compatible with convergence of F(N)/N to a positive limit <= 2/3; neither existence nor positivity of the limit is provable by these means.

Additional exact facts within the completely stored range [1, 10^6] (F = 535,585) [COMPUTED]: first elements 2, 3, 5, 8, 9, 14, 15, 17, 23, 24, 26, ...; smallest non-members 1, 4, 6, 7, 10, 11, 12, 13, 16, 18, ...; maximal gap between consecutive elements is 10 (2159 to 2169); conditional density inside allowed classes {n not 1 mod 3} rises monotonically: 0.591, 0.634, 0.721, 0.773, 0.803 at N = 10^2, ..., 10^6.

7. Mean-field heuristic: derivation and empirical rejection

Membership is an OR over divisor pairs (Theorem 1): n is in A iff n+1 = ab for some a,b in A (or n in {2,3}). Assume **(H1)** equidistribution: for each b, |A ^ [1,M] ^ (-1 mod b)| ~ F(M)/b uniformly in b. Counting candidate images ba - 1 <= N over all b in A gives F(N) ~ sum_{b in A} F(N/b)/b. Inserting the ansatz F(N) ~ c N^alpha and cancelling yields the self-consistency equation **(*)**: sum_{b in A} b^-(1+alpha) = 1. Because the truncated harmonic sum diverges like delta_hat ln X (computed H_X = 9.1036 at X = 10^8, increments approaching 1.287 per decade), the left side decreases continuously from infinity to zero, so under (H1) a unique alpha* > 0 exists **[HEURISTIC]**.

Numerics [COMPUTED]. Exact truncated sums below 2*10^5 and 64 logarithmic buckets per decade above give alpha*_trunc(10^8) = 0.47344208837849144 (root residual +2.5e-11; stable: 0.472709 already at X = 10^6). The integral surrogate delta 2^-alpha / alpha = 1 gives alpha*(delta = 0.5594) = 0.4185 and alpha*(delta = 2/3) = 0.4785.

Rejection [COMPUTED vs HEURISTIC]. The mean field predicts F(10^8) ~ (10^8)^0.4734 = 6,131 against the exact 55,939,931: off by factor 9,124. Simultaneously the direct fit sits at alpha_OLS = 1.016 > 1 > alpha*. Two mechanisms identify the failure: (1) *self-affine imaging* - each b in A transports a near-copy {ba - 1 : a in A} into A, so membership correlations inflate density beyond products of marginals (the OR-condition over divisor pairs); (2) *modular rigidity* - equidistribution fails absolutely on class 1 mod 3, and Proposition 3 formalizes such constraints for every modulus. Equation (*) therefore describes a random model of A, not A itself. Its interest is negative but sharp: GREEN-036 is not governed by pseudorandom multiplicative statistics, and any successful approach must exploit the correlated, self-affine structure.

8. Outlook: what is missing

[OPEN] To prove $d(A) > 0$ one wants $F(N) >= cN$, plausibly via eventual containment of an arithmetic progression of allowed class, or a bootstrap converting lower density at height T into higher density at height T' > T. Visible obstructions: 6, 11, 62 are absent despite allowed residues and the gap 2159-2169 shows early holes, so any progression claim needs a threshold phenomenon. To prove $d(A) = 0$ one must reverse a trend rising across eight decades of exact computation; no mechanism suggests itself. Between the two: the only proven ceiling is 2/3 (Lemma A), the residue method is exhausted for $m <= 500$, and the proven growth floor is $(\ln N)^2/(4 \ln^2 2)$. Any genuinely new constraint - larger-modulus rigidity, p-adic restrictions, complement stability - would constitute real progress on GREEN-036.

Reproducibility. Everything regenerates by python3 verify_solution.py [X] (default X = 10^7, about 20 s; production X = 10^8, 6.5 min, nine of nine checks passed). Artifacts: data/census_summary.json, data/census.csv.gz (first 10^6 elements), data/run_output.txt reproduced verbatim in the Appendix. Standard library only; deterministic.

Appendix. Verbatim verifier output (production run, X = 10^8)

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GREEN-036 verification run, X = 100000000
======================================================================
[1] exact census DP: |A cap [1,100000000]| = 55939931   (351.06s)
[2] naive closure cross-check at X=3000: DP |A|=1358, naive |A|=1358, identical=True   (0.57s)
[3] Lemma A: elements congruent to 1 (mod 3) in census: 0 (out of 55939931); predicted 0
    observed raw density F(100000000)/100000000 = 0.559399  (proven upper bound 2/3 = 0.666667)
[4] count table |A cap [1,10^k]| and per-decade increments (25.78s):
      k   F(10^k)      increment   F/N       local exp  harm incr  delta_hat
      1          5            5   0.500000       nan    1.2694  0.5513
      2         39           34   0.390000    0.8921    0.8178  0.3552
      3        422          383   0.422000    1.0342    0.9470  0.4113
      4       4805         4383   0.480500    1.0564    1.1029  0.4790
      5      51508        46703   0.515080    1.0302    1.1838  0.5141
      6     535585       484077   0.535585    1.0170    1.2318  0.5350
      7    5493428      4957843   0.549343    1.0110    1.2637  0.5488
      8   55939931     50446503   0.559399    1.0079    1.2871  0.5590
[5] Lemma B lower-bound families:
    P-orbit 2^k+1, k=1..26: 26 values, all in A: True
    Z-products (2^i+1)(2^j+1)-1 <= X (i<=j): generated 168, in A: 168, collisions: 0
    proven counting formula floor((floor(log2 X)-1)^2/4) = 156 (<= actual 168)
    lower bound achieved: 168 = 3.00322e-06 x F(100000000)
    mixed (2^i+1)((3^j+1)/2)-1 <= X: generated 216, in A: 216, collisions: 0
    asymptotic constants: pure-2 (PROVEN) 1/(4 ln^2 2) = 0.52034, mixed (if injective) 1/(2 ln2 ln3) = 0.65660
    (0.00s)
[6] residue census, 95 primes <= 500 (6.46s):
    primes with PROPER S_p: [(3, 2)]
    composite moduli 4..500 beating 2/3: none
    S_3 = [0, 2] (Lemma A), |S_p| = p for all other primes p <= 500: True
    combined CRT bound over proper prime constraints: 0.666667 (= proven 2/3 here); best single 0.666667
[7] growth-exponent estimates:
    OLS log-log fit over k=4..8: alpha_hat = 1.01600 (R^2 = 0.999981)
    top local exponent ln(F_8/F_7)/ln 10 = 1.00788
    truncated harmonic sum H_X = sum_(b in A cap [1,100000000]) 1/b = 9.103598
    truncated zeta grid: S(0.0)=9.1039, S(0.25)=1.8511, S(0.5)=0.9483, S(0.75)=0.6321, S(1.0)=0.4615
    solved S_X(alpha*) = 1: alpha*_trunc = 0.473442 (residual +2.51e-11)
    mean-field prediction F(100000000) ~ X^alpha* = 6131 vs actual 55939931  (ratio 9124)
[8] artifacts written in 0.9s: census_summary.json, census.csv.gz (first 1000000 elements) sizes: 3618B, 2285592B
======================================================================
[PASS] EXACTNESS      DP set == naive fixpoint closure on [1,3000] (|A| = 1358 both sides: True)
[PASS] LEMMA-A        0 violations of n ≡ 1 (mod 3) among 55939931 census elements
[PASS] PORBIT         all 26 values 2^k+1 <= X lie in A
[PASS] LEMMA-B-Z      168/168 Z-products in A, 0 collisions, proven formula 156 <= actual
[PASS] LEMMA-B-MIXED  216/216 mixed products in A, 0 collisions (injectivity COMPUTED, not proven)
[PASS] RESIDUES       S_3 = {0,2}; proper-prime list [(3, 2)]; combined bound 0.666667 >= observed density 0.559399
[PASS] MONOTONE       decade counts strictly increasing, e.g. 5 -> 55939931
[PASS] ZETA-ROOT      S_X(alpha*) = 1 at alpha* = 0.473442, residual +2.51e-11
[PASS] DENSITY-TREND  F/N still rising: 0.5493 -> 0.5594 (question OPEN)
----------------------------------------------------------------------
9/9 checks passed.  Problem GREEN-036 remains OPEN; this run establishes computational evidence only.
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